

| UCE513: CONSTRUCTION MANAGEMENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |     |
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| <p><b>Course Objective:</b> This subject aims to develop an understanding of principles and techniques of estimating construction costs, with emphasis on quantity take-off and pricing elements of work. It also covers the basic principles, techniques, and practices used as management tools by contractors to plan, schedule, and control time and costs on building various infrastructure projects.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |     |
| <p><b>Quantity Surveying and Cost Estimation :</b>Quantity survey measurements, Bill of quantities, analysis of rates for different items of work, Specifications and Units of different items of work, Detailed Estimate of various items of Building and Pavements using MS Excel, Common schedule of rates for different items of works, Bar Bending Schedule</p> <p><b>Contracts:</b> Types of contracts and their characteristics, procedure for tendering/e-tendering and contracts, BOT, DBFOT, PPP &amp; HAM financial models, evaluation and examination of tenders, award of work, Joint Ventures, Concession Agreements,</p> <p><b>Network Techniques:</b> Methods of Scheduling-Gantt Chart, Bar chart, Development of Bar charts and Gantt chart, Merits &amp; limitations of Bar chart &amp; Gantt chart. Concept of CPM &amp; PERT: Introduction to Critical path method (CPM), Program evaluation &amp; review techniques (PERT), Network Diagramming of Projects Activity-on arrow (AOA) Diagrams- Concept of Activity and Event, Time-Analysis of Networks- Forward Pass, Backward Pass, Probabilistic Durations- Optimistic Time, Pessimistic Time, Most Likely Time, Project Scheduling- ES and LS Schedules as Limits, Resource Scheduling, Time/Cost Trade-off Definitions, functions &amp; characteristics of project planning and principles of project Planning and Management, Bar milestone charts, Planning and scheduling with PERT / CPM, Time cost optimization, Probability concepts Allocation of resources and resource levelling, Updating, Application of software for Project planning, scheduling</p> |   |   |   |     |
| <p><b>Experimental Project/assignment/Micro Project</b></p> <ol style="list-style-type: none"> <li>1. Complete cost estimation of the building or highway project using MS Excel</li> <li>2. Development of a contract document for the infrastructure project.</li> <li>3. Planning &amp; Scheduling of infrastructure project using software.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |     |
| <p><b>Course Learning Objectives (CLO)</b></p> <p>Upon completion of this course, the students will be able to:</p> <ol style="list-style-type: none"> <li>1. Perform the rate analysis for the various construction activities</li> <li>2. Estimate the cost for the building and the road projects</li> <li>3. Perform the project planning, scheduling, and time-cost optimization of the project.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |     |

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| 4. Perform resource allocation and project controlling of the project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Text Books:</b> <ol style="list-style-type: none"> <li>1. Kohli D. S.; A Text book on Estimating and Costing and Accounts, S. Chand &amp; Company New Delhi (2013)</li> <li>2. R.L. Peurifoy, W.B.Ledbetter and C.J.Schexnayder,“ Construction planning and methods”, Fifth edition, McGraw Hill International edition(2010)</li> </ol><br><b>Reference Books:</b> <ol style="list-style-type: none"> <li>1. Seetharaman S., Construction Engineering and Management,Umesh Publication Delhi(2017)</li> <li>2. Punima B. C. and Khandelwal; Project Planning and Control with PERT and CPM, Laxmi Publication New Delhi(2016)</li> <li>3. K.K. Chitkara, Construction project management: planning, scheduling and controlling, Tata McGraw-Hill.(2014)</li> <li>4. L.S. Srinath, PERT and CPM Principles and Application, Third edition, Affiliated east-west press Pvt Ltd (2001)</li> </ol> |

#### Evaluation Scheme:

| Sr. No. | Evaluation Elements                                                             | Weights (%) |
|---------|---------------------------------------------------------------------------------|-------------|
| 1.      | MST                                                                             | 25          |
| 2.      | EST                                                                             | 35          |
| 3.      | Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations) | 40          |